Practical no. 4

Write a program in C++ to perform following operations on complex numbers Add, Subtract, Multiply, Divide, Complex conjugate. Design the class for complex number representation and the operations to be performed. The objective of this assignment is to learn the concepts classes and objects

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Program :

```
#include <iostream>
using namespace std;

class complex
{
    float real;
    float imaginary;

public:
    void getData()
    {
        cout << "Enter real number: ";
        cin >> real;

        cout << "Enter imaginary number: ";
        cin >> imaginary;
    }

    void displayData()
    {
        if (imaginary < 0)
        {
            cout << real << imaginary << "i" << endl;
        }
        else
```

```
    {
        cout << real << "+" << imaginary << "i" << endl;
    }
}

complex add(complex c)
{
    complex temp;

    temp.real = real + c.real;
    temp.imaginary = imaginary + c.imaginary;

    return temp;
}

complex subtract(complex c)
{
    complex temp;

    temp.real = real - c.real;
    temp.imaginary = imaginary - c.imaginary;

    return temp;
}

complex multiply(complex c)
{
    complex temp;

    temp.real = (real * c.real) - (imaginary * c.imaginary);
    temp.imaginary = (real * c.imaginary) + (imaginary * c.real);

    return temp;
}

complex divide(complex c)
{
    complex temp;

    float denominator;
    denominator = (c.real * c.real) +
                  (c.imaginary * c.imaginary);

    if (denominator == 0)
```

```
        {
            cout << "Error: Cannot divide by zero complex number."
                 << endl;

            temp.real = 0;
            temp.imaginary = 0;
            return temp;
        }

        temp.real = ((real * c.real) +
                     (imaginary * c.imaginary)) / denominator;

        temp.imaginary = ((imaginary * c.real) -
                          (real * c.imaginary)) / denominator;

        return temp;
    }
};

int main()
{
    complex s1, s2, s3, s4, s5, s6;

    s1.getData();
    s2.getData();

    cout << "\nFirst Complex Number: ";
    s1.displayData();

    cout << "Second Complex Number: ";
    s2.displayData();
    cout << "\n";

    s3 = s1.add(s2);

    cout << "Addition: ";
    s3.displayData();

    s4 = s1.subtract(s2);

    cout << "Subtraction: ";
    s4.displayData();

    s5 = s1.multiply(s2);
```

```
    cout << "Multiplication: ";
    s5.displayData();

    s6 = s1.divide(s2);

    cout << "Division: ";
    s6.displayData();

    return 0;
}
```

```
Enter real number: 4
Enter imaginary number: 2
Enter real number: 6
Enter imaginary number: 3

First Complex Number: 4+2i
Second Complex Number: 6+3i

Addition: 10+5i
Subtraction: -2-1i
Multiplication: 18+24i
Division: 0.666667+0i


...Program finished with exit code 0
Press ENTER to exit console.
```